

SHREYASHI

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EDUCATION

Vellore Institute of Technology, Vellore — 2027

B. Tech, Electronics and Communication Engineering | CGPA: 9.1

School of Creative Learning, Patna, Bihar — 2023

CBSE (Class XII) | Percentage: 93%

Notre Dame Academy, Patna, Bihar — 2021

CBSE (Class X) | Percentage: 95.6%

EXPERIENCE

Kineton (VITTBI Startup) | Feb 2025 – Dec 2025

- Designed pipelined architectures, FSMs, combinational and sequential logic and created synthesizable RTL modules in Verilog.
- Power, performance, and area (PPA) optimized critical paths.
- Helped the physical design and verification teams with debugging.

L&T Edu Tech: Industrial IoT Summer Virtual Internship | May 2025–July 2025

- Acquired hands-on experience with edge devices, smart sensors, and IIoT architecture.
- Developed industrial protocols for data exchange, such as Ethernet, MQTT, and Modbus.
- Investigated IoT applications in cloud platforms, automation, and energy management.

ACHIEVEMENTS

1. Integrated Smart Charging System (PATENT PUBLISHED)

30 June 2025

Application no:202541062114

2. Non-invasive Smart Helmet with Integrated Biometric Fatigue Detection (PATENT PUBLISHED)

30 Sept 2025

Application no:202541093702

3. Winner of ARC Mobility (Smart Mobility and Intelligent Systems) Hackathon 2026

SKILLS

Programming Languages: C, Embedded C, Java (Basics), Assembly (8051)

Hardware & Embedded Development Tools: Keil µVision, Arduino IDE, Tinker Cad, Multisim, Quartus Prime, Blynk IoT, ThingSpeak

Simulation & Modeling Software: MATLAB, Simulink, AWR, Cadence, HFSS

Hardware: Microcontrollers (Arduino, ESP32, STM32), Sensor Integration, Wireless Communication (RF module), communication protocols (I2C, UART, SPI, HTTP, Modbus, Ethernet), Analog circuit Design (Op-Amps, Filters, Amplifiers), Transistor Level Design (BJT, MOSFET), Analog Signal Processing, Hardware Testing (Function Generator, Oscilloscope), FPGA Boards

Soft skills: Leadership, Teamwork, efficient Time Management, Creativity & Innovation, Adaptability & Quick Learning

PROJECTS

1. REAL-TIME HARDWARE LEVEL ANOMALY DETECTION USING SIGNAL SHAPE ANALYSIS

Feb 2026

Designed and implemented a real-time embedded system for hardware-level anomaly detection, analysing signal waveform characteristics and detecting deviations using signal shape feature extraction techniques.

2. VISION-LANGUAGE ACCESSIBILITY AI MODEL IN INDIAN LANGUAGES

July 2025

A specialised AI system that interprets images and generates captions in Indian languages.

It can describe visual content, translate text from images, and help visually impaired people by converting visual data into audio.

3. AUDIO SPECTRUM ANALYSER USING ADAPTIVE FILTER

April 2025

Captured real-time sound using a microphone and performed FFT and PSD analysis in MATLAB with DC offset removal and noise suppression using LMS and moving average filters.

4. EDGE AI MODEL SIMULATION USING RASPBERRY PI

March 2025

Built a simulated AI inference interface on the Wokwi Raspberry Pi Pico, demonstrating MNIST digit recognition using a lightweight TensorFlow Lite model. Showcased how Edge AI can deliver real-time predictions even in resource-constrained environments.

5. SMART HELMET

March 2025

Developed a Smart Helmet system that verifies helmet use, detects alcohol levels, and sends GPS location in case of crashes—automating safety checks before ignition. Integrated MQ-3, IR, SW-420, GPS, and NRF24L01 modules with Arduino to enhance rider safety through wireless control.

6. INTEGRATED SMART BATTERY CHARGER

October 2024

Designed a hybrid analog-digital system to boost battery life, safety, and user control for small electronic devices, featuring timers, alarms, and charge time estimation. Enabled USB-based excess energy transfer to secondary devices, enhancing energy efficiency and parental control.

7. GREENHOUSE AUTOMATION SYSTEM

January 2024

Developed a smart monitoring system that enhances greenhouse efficiency to optimize plant growth using temperature, humidity, pressure, and obstacle detection sensors.

EVENTS AND PARTICIPATIONS

1. Hackathons

Participated in Fusion 360 Hackathon by Robovitics.

Finalist in Yantra Electronics Hackathon by IEEE SPS Club

Participated in Yantra Central Hackathon'25

2. Workshops

iAntenna'25 – Intelligent Antenna Design Contest & Workshop by Industry Academia Conclave'25

3. Clubs & Volunteering

Student Coordinator (STALLS RIVIERA'26)

Core member in Robovitics Club

Member of BIF Club (NGO Club)